Functions of Parathormone

The parathormone regulates calcium and phosphorus metabolism in the body.They act on bone, kidney and intestine to maintain blood calcium levels.

Parthyroid Dysfunction

The secretion of parathyroid hormone can be altered due to the following conditions.

Removal of parathyroid glands during thyroidectomy (removal of thyroid) causes decreased secretion of parathormone. The conditions are

- Muscle spasm known as **Tetany** (sustained contraction of muscles in face, larynx, hands and feet).
- Painful cramps of the limb muscles.

16.2.4 Pancreas (Islets of Langerhans)

Pancreas is an elongated, yellowish gland situated in the loop of stomach and duodenum. It is **exocrine** and **endocrine** in nature. The exocrine pancreas secretes pancreatic juice which plays a role in digestion while, the endocrine portion is made up of Islets of Langerhans.

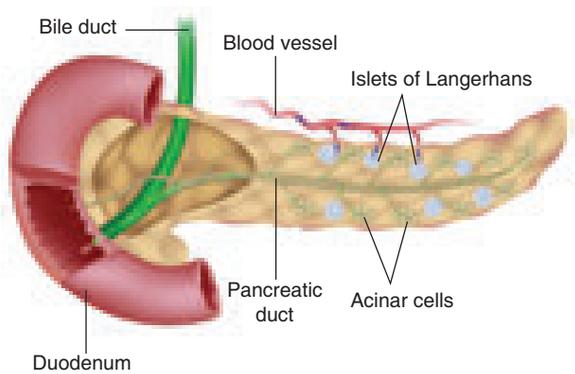


Figure 16.10 Pancreas



Human insulin was first discovered by Fredrick Banting, Charles Best and MacLeod in 1921. Insulin was first used in treatment of diabetes on 11th January 1922.

The Islets of Langerhans consists of two types of cells namely alpha cells and beta cells. The **alpha cells** secrete **glucagon** and **beta cells** secrete **insulin**.

Functions of Pancreatic hormones

A balance between insulin and glucagon production is necessary to maintain blood glucose concentration.

Insulin

- Insulin helps in the conversion of glucose into glycogen which is stored in liver and skeletal muscles.
- It promotes the transport of glucose into the cells.
- It decreases the concentration of glucose in blood.

Glucagon

- Glucagon helps in the breakdown of glycogen to glucose in the liver.
- It increases blood glucose levels.

Diabetes mellitus

The deficiency of insulin causes **Diabetes mellitus**. It is characterised by

- Increase in blood sugar level (Hyperglycemia).
- Excretion of excess glucose in the urine (Glycosuria).
- Frequent urination (Polyuria).
- Increased thirst (Polydipsia).
- Increase in appetite (Polyphagia).

16.2.5 Adrenal Gland

The adrenal glands are located above each kidney. They are also called **supra renal glands**.

The outer part is the adrenal cortex and the inner part is the adrenal medulla. The two distinct parts are structurally and functionally different.

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